



Safe Routes to School

Student team: Doris Gao, Evelyn Lui, Lillian Zhao

Client: City of Pittsburgh – Harriet Jackson

Team Introduction



Evelyn Lui

Student Team



Lillian Zhao

Student Team



Doris Gao

Student Team



Harriet Jackson

Community Partner



Professor Shihong

Faculty Advisor

A black and white photograph of a park path. In the foreground, a low stone curb separates a grassy area from a paved path. The path leads into the distance, flanked by trees. The background is blurred, showing more trees and what appears to be a street with some lights. The word "Introduction" is overlaid in the center in a bright yellow, bold, sans-serif font.

Introduction



Our Community Partner

Safe Routes to School (SRTS) program operates within City of Pittsburgh DOMI run by Program Coordinator Harriet Jackson

Works and coordinates with many other DOMI departments such as traffic, city planning, public works, public safety, etc.

Serves Schools in the City of Pittsburgh

- Student safety through infrastructure improvements for walking and biking routes as well as traffic safety.





Problem: Walkability Audits



What to Look For list:



Examples Of What To Look For:

- No sidewalk or trail
- Too narrow
- Blocked/cluttered
- Not Maintained
- No Pedestrian signal
- Walk Indicators are not audible
- Timing of signals inadequate
- Absence of level landings
- Faded or illegible signage
- No texture/color change at intersection
- Not parallel to the traffic flow
- Absence of curb ramps
- View of oncoming traffic is blocked

Kittichulab Department of Mobility and Infrastructure

8/8/2022

Manual Data Entry

- Physical clipboards requires manually transcriptions
- Introduces errors and costs hours

No Audit History

- Past findings scattered across files, emails, and physical storage
- Which schools have recurring problems? Are conditions improving?

No Systematic Report Mechanism

- Compiling findings for each audience group takes days
- Difficult to respond promptly to stakeholders



Solution

 Pre/Post surveys (Survey123)



 Data Collection/Cleaning (Feature Layer)



 Custom LLM Report Generation Web-app

Minimize
disruption
of current
workflow

Plan for
future use
and
maintenance

Integrated in
ArcGIS
framework





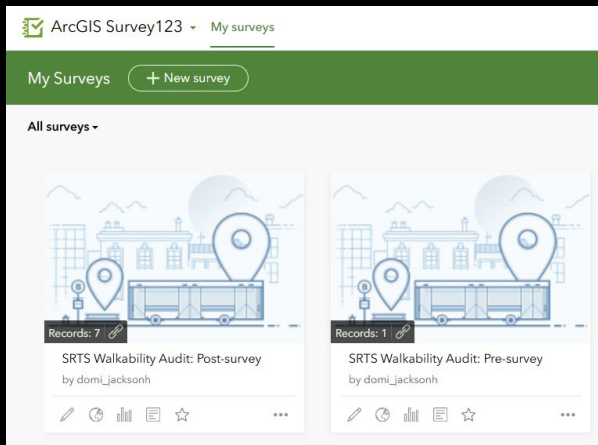
Solution

Pre/Post surveys
(Survey123)

Data
Collection/Cleaning
(Feature Layer)

Custom LLM
Report Generation
Web-app

- **Mobile** integrated
- Divided by **roles**



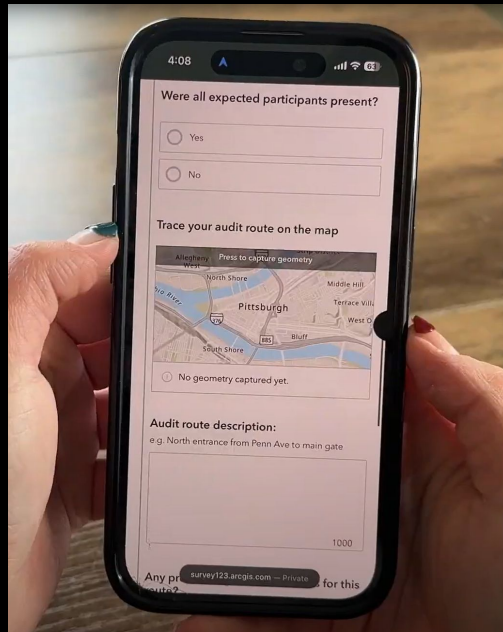


Solution

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Includes **Geotracing** of Audit Route

EditDate	What is your full name	What is your email address	What is your role	Date of audit	Weather Conditions	Other
4/23/26, 5:23 PM	George Barrow	george.barrow@pittsburg...	Planner	March 26, 2026	Clear	
4/23/26, 2:41 PM	Evelyn Lui	evelynlui@andrew.cmu.edu	Guest	April 23, 2026		
4/22/26, 5:21 PM	Harriet Jackson	harriet.jackson@pittsburg...	SRTS Coordinator	March 26, 2026	Clear	
3/31/26, 1:46 PM	George Barrow	George.barrow@pittsburg...	Traffic	March 26, 2026	Clear	
3/27/26, 5:54 AM	Tanika Huffman	Thuffman1@pghschool.org	Guest	March 26, 2026	Clear	
3/26/26, 10:42 AM	Yuhan Gao	yuhang2@andrew.cmu.edu	Guest	March 26, 2026	Clear	
3/26/26, 10:10 AM	Graham Mulqueen	Gmulqueen@pghschool...	Guest	March 26, 2026	Clear	



Solution

 Pre/Post surveys
(Survey123)



 Data
Collection/Cleaning
(Feature Layer)



 Custom LLM
Report Generation
Web-app

SRTS Report Builder
Safe Routes to School - Pittsburgh

Signed In Sign out

HOW IT WORKS

1  **Sign In**
Authenticate with your ArcGIS Pittsburgh account to access survey data.

2  **Select an Audit**
Choose the walkability audit you want to generate a report for.

3  **Preview Data**
Review the survey responses that will be used to generate your report.

4  **Download Report**
Generate and download an AI-written report tailored to your audience.

1 **Sign In** Signed in

You're signed in. Session valid until 11:11:42 AM.

✓ 2 audits loaded — scroll down to select one.

2 **Select an Audit** Refresh Audit selected

2 audits loaded

Choose an audit

PITTSBURGH MORROW PREK-8 - INTERMEDIATE — April 23, 2▼

✓ Selected — scroll down to preview your data.

3 **Preview Survey Data** Load Preview

Verify the data that will be used to generate your report.

Click Load Preview to fetch and display the survey responses for this audit.

4 **Download Report**



Claude

- Simple **authentication** pipeline
- **Automated** audit sort
- **Preview** data

4 **Download Report**

Choose a report type below. Each is written by AI using the audit data and tailored for a different audience. Reports download as a .docx file.

DOMI Internal Confidential

Detailed technical findings for city planners and DOMI staff. Includes all data, severity ratings, and infrastructure notes.

Download

School & Community For schools

Accessible summary for school administrators, PTAs, and community members focused on student safety.

Download

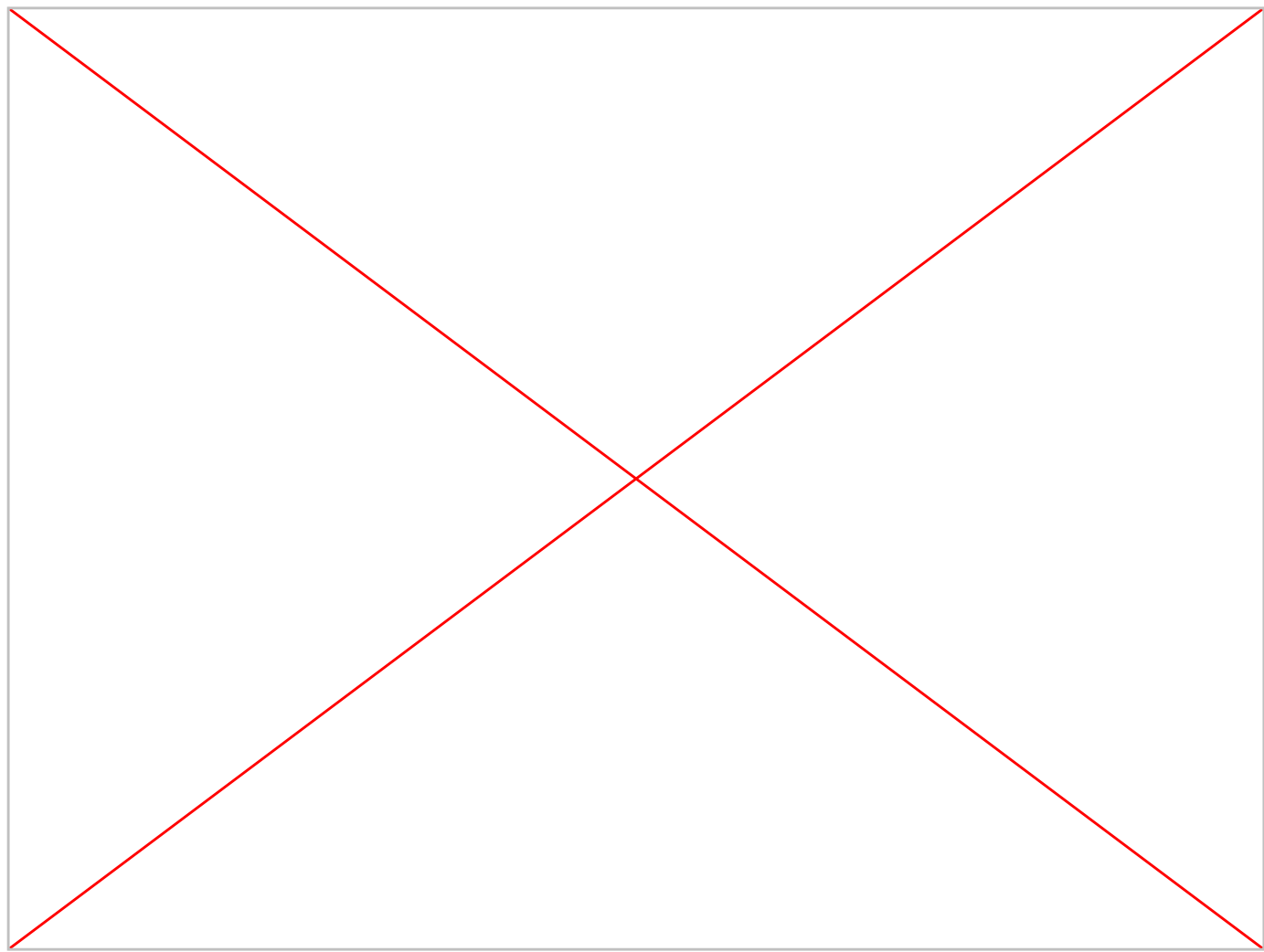
Public Update Public

Plain-language overview suitable for public newsletters, social media, or community meetings.

Download

A black and white photograph of a park path. In the foreground, a low stone wall runs diagonally across the frame, with some grass growing behind it. The path continues into the distance, flanked by trees and foliage. The background is softly blurred, showing more trees and what appears to be a street with some light sources. The overall mood is peaceful and natural.

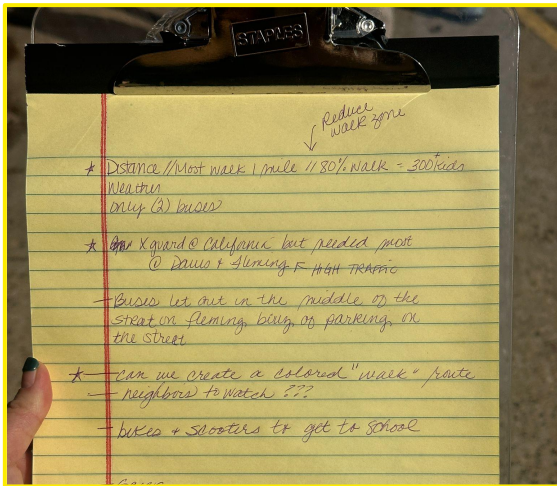
Outcomes/Impact



Expanded Capacity



Phase 1: Audit



Phase 2: Survey

CITY OF [PITTSBURGH](#) DEPARTMENT OF MOBILITY & [INFRASTRUCTURE](#) SAFE ROUTES TO SCHOOL

Safe Routes to School — Community Update
PITTSBURGH MORROW PREK-8 - [PRIMARY](#) | March 26, 2026 | Public

SAFE ROUTES TO SCHOOL

Walkability Audit — Community Update
PITTSBURGH MORROW PREK-8 - [PRIMARY](#) | March 26, 2026

About the Program

Safe Routes to School is a City of Pittsburgh program dedicated to making it safer and easier for children to walk and bike to school. Walkability audits bring together school staff, city officials, and community members to walk the routes students take and identify ways to improve safety.

This Audit at a Glance

PITTSBURGH MORROW PREK-8 - PRIMARY School Audited	51-75% Students Walk to School	High Overall Severity
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Route Walked

We walked from the school on Fleming Ave, to Davis Ave making a right on Brighton Rd, walked down Brighton Rd, turning left on Woods Run Ave, [Right](#) on McClure Ave, to Antrim St. back to the school on Fleming Ave.

Phase 3: Report



How we surveyed before...



Moving forward...

- Ability to do more audits
- Getting/receiving information faster from the audits
- Sharing audit reports faster to the different participants, partners, and colleagues

>> Resolving more safety issues and concerns